

KEYSIGHT

TECHNOLOGIES

CloudLens Stack

Deployment Runbook

CLMS, vPB and sensors. End to end. One command.

Version	v1.0
Date	June 2026

Keysight Technologies | Network Visibility Solutions

Table of Contents

- 1. Executive Summary.....
- 2. The Three Deployment Paths.....
- 3. Prerequisites.....
- 4. Path A: Bash One-Liner.....
 - 4.1 What the script does (phase by phase).....
 - 4.2 Flags and overrides.....
- 5. Path B: Terraform Stack Module.....
 - 5.1 Workflow.....
 - 5.2 What gets created.....
- 6. Path C: Azure Portal.....
- 7. Verification Checklist.....
- 8. Troubleshooting Reference.....
- 9. Cleanup / Decommission.....
- 10. Appendix: File Paths and Quick Links.....

1. Executive Summary

This runbook shows how to deploy the full CloudLens stack on Azure in one shot: CloudLens Manager (CLMS), optional Virtual Packet Broker (vPB), and the OS sensors that connect tagged VMs back to CLMS. Three paths are documented; pick the one that matches how your team works. All three produce the same Azure resources.

Audience

- Customer DevOps and platform engineers who own the Azure subscription.
- Keysight Sales Engineers running customer POCs or demos.
- Customer CTO / procurement / training teams who need a printable reference.

What gets deployed

Component	Azure resources	Required
CloudLens Manager (CLMS)	1 VM (D4s_v5), 1 public IP, NSG, NIC	Yes
Virtual Packet Broker (vPB)	1 VM (D8s_v3), 1 public IP for mgmt, NSG, 3 NICs	Optional
Sensors	Agent containers (Linux) and native service (Windows) on tagged VMs	Optional
Shared networking	VNet 10.50.0.0/16, 1-4 subnets depending on options	Yes

Time and cost

Phase	Wall-clock time
CLMS VM provisioned and initialised	15 to 20 min
vPB VM provisioned (optional)	5 to 10 min
Sensors deployed on first 10 VMs (optional)	5 to 8 min
Total typical (CLMS + vPB + sensors)	30 to 40 min

Indicative Azure cost (East US 2, June 2026 list prices): CLMS D4s_v5 about USD 140/month, vPB D8s_v3 about USD 280/month, plus small storage and public-IP fees. Stopping the VMs deallocates compute charges. Marketplace software fees are billed separately by Keysight.

2. The Three Deployment Paths

All three paths accept Azure Marketplace terms automatically, deploy the same VMs, and chain CLMS first, then vPB. Pick the one that matches your operating model. You can switch between them later without losing state, because each path produces standard Azure resources in a resource group.

Path	Best for	Tools needed	Time to first paste
A. Bash one-liner	First-time trial, customer demo, Cloud Shell users	Browser + Azure CLI (Cloud Shell has both)	10 seconds
B. Terraform stack module	IaC pipelines, repeatable customer envs, GitOps	Terraform 1.5+ and Azure CLI logged in	2 minutes
C. Azure Portal	Click-through customers, governance-controlled subscriptions	Browser only	1 minute

Why three?

Customers do not all work the same way. A platform team wants Terraform. A network engineer trialling the kit wants a paste. A CTO walking through with procurement wants buttons. Three paths, identical outcome.

3. Prerequisites

Confirm each item before kicking off the deployment. The script will stop early with a clear message if any of the first three are missing, so you do not waste time mid-deploy.

- ☐ Azure subscription with Contributor role on the target resource group (or rights to create one).
- ☐ Azure CLI 2.50 or later installed and logged in. In Cloud Shell this is preinstalled and pre-authenticated.
- ☐ Marketplace terms can be accepted automatically by the script; you do not need to click through anything in the Portal first.
- ☐ Quota for the DSv5 family (CLMS, 4 vCPU) and DSv3 family (vPB, 8 vCPU) in your chosen region. The script does a soft quota probe and warns without blocking.
- ☐ Open egress on the VM subnets to reach the CLMS public IP on TCP/443 (only needed once sensors are deployed).

Region note

Default region is eastus2. CLMS and vPB Marketplace images are also published in westus2, northeurope, westeurope, southeasia. Run 'az vm image list --offer keysight-cloudlens-manager-preview --publisher keysight-technologies-cloudlens --all' to confirm before trying a less-common region.

4. Path A: Bash One-Liner

The fastest way to a working stack. One paste in Azure Cloud Shell or any local terminal with the Azure CLI; the script handles the rest.

```
curl -sSL https://raw.githubusercontent.com/Keysight-Tech/cloudlens-ansible-azure/main/deploy/deploy-stack.sh | bash
```

Prefer to inspect before running? Download then execute:

```
curl -sSL -o deploy-stack.sh \  
https://raw.githubusercontent.com/Keysight-Tech/cloudlens-ansible-azure/main/  
deploy/deploy-stack.sh  
less deploy-stack.sh  
bash deploy-stack.sh
```

4.1 What the script does (phase by phase)

The script runs 10 phases. Each phase prints a banner with its name, so if anything fails you know exactly where you are.

1. Banner and environment detection: prints the version header and detects Cloud Shell vs local terminal so prompts adapt accordingly.
2. Pre-flight checks: confirms az CLI is installed, you are logged in, and probes vCPU quota for the DSv5 and DSv3 families.
3. Customer input: prompts for resource group name (default cloudlens-rg), Azure region (default eastus2), and a strong admin password. All defaults can be overridden by flags (see 4.2).
4. Marketplace terms acceptance: runs 'az vm image terms accept' for the CLMS and (if selected) vPB plans so the deployments do not stall on marketplace consent.
5. Resource group creation: creates the RG only if it does not exist. Re-running the script against an existing RG is safe.
6. CLMS deployment: deploys the clms-marketplace.json ARM template. Waits for the VM provisioning state to be Succeeded and captures the public IP.
7. Wait for CLMS init: polls the CLMS HTTPS endpoint until the UI is reachable (typically 15 minutes). Prints a one-line progress bar.
8. vPB deployment (optional): the prompt defaults to Yes; type N to skip. When enabled, deploys vpb-marketplace.json and captures the management public IP.
9. Manual project key step: the script pauses and tells you to open the CLMS UI, create a project, and paste the project key back into the terminal. This is the one step that cannot be automated end to end today (CLMS does not yet expose a project-create REST endpoint).
10. Sensor chain (optional): hands off to quickstart.sh, which deploys sensors to every Azure VM tagged cloudlens=yes. Prints a final summary and writes cloudlens-deploy-summary.txt with all IPs and credentials.

Output you can keep

Every run writes 'cloudlens-deploy-stack.log' (raw stdout/stderr) and 'cloudlens-deploy-summary.txt' (RG name, VM names, IPs, default creds). Save both to your customer ticket or runbook archive.

4.2 Flags and overrides

Flag	Effect
--dry-run	Walk through every prompt and print the az commands without touching Azure. Use this to preview the run.
--resource-group NAME	Override the default RG name (cloudlens-rg). Useful when deploying into an existing customer RG.
--location REGION	Override the default region (eastus2). Pass any region that carries the Marketplace images.
--no-vpb	Skip the vPB phase entirely. CLMS-only deploys complete in about 20 minutes.
--no-sensors	Stop after the CLMS (and vPB) phase. Run quickstart.sh later yourself if you want sensors then.
-h --help	Print the help banner with all flags and a phase summary.

5. Path B: Terraform Stack Module

The stack module wraps the existing clms and vpb modules behind one tfvars file. One 'terraform apply' provisions the resource group, a shared VNet with the right subnets, both VMs, NSGs, NICs and public IPs. The sensors are still deployed separately via Ansible because they touch customer VMs, not Azure resources.

5.1 Workflow

11. Clone the repository and change into the stack module directory.
12. Copy terraform.tfvars.example to terraform.tfvars and fill in subscription_id and admin_password.
Toggle deploy_vpb if you only want CLMS.
13. Run terraform init to download the azure provider.
14. Run terraform plan to review the resources Terraform will create. Confirm against your subscription's policies.
15. Run terraform apply. About 8 to 15 minutes later, CLMS and vPB are ready. The outputs print the CLMS UI URL, the vPB management IP and the default credentials.
16. Open the CLMS UI, create your project, copy the project key, and run quickstart.sh (or deploy-stack.sh --no-vpb) to deploy sensors.

```
git clone https://github.com/Keysight-Tech/cloudlens-ansible-azure.git
cd cloudlens-ansible-azure/deploy/terraform/stack
cp terraform.tfvars.example terraform.tfvars
EDITOR terraform.tfvars # set subscription_id, admin_password
terraform init
terraform plan
terraform apply
```

5.2 What gets created

Resource	Count	Notes
Resource group	1	cloudlens-rg (override via tfvars)
Virtual network	1	cloudlens-stack-vnet 10.50.0.0/16
Subnets	4	clms-subnet, vpb-mgmt, vpb-ingress, vpb-egress
Public IPs	2	One for CLMS, one for vPB management
NSGs	2	Pre-configured rules per VM role
NICs	4	1 for CLMS, 3 for vPB
VMs	2	CLMS D4s_v5, vPB D8s_v3

Existing VNet?

Set `shared_vnet = false` in `terraform.tfvars`. Each child module then creates its own VNet (CLMS gets a 1-subnet VNet, vPB gets a 3-subnet VNet). For full bring-your-own-VNet wiring, edit `main.tf` and pass `existing_vnet_name` through to the child modules.

6. Path C: Azure Portal

If you prefer to deploy from the Portal, the same Marketplace images are also exposed via the two click-through buttons on the landing page at keysight-tech.github.io/cloudlens-ansible-azure. The Portal opens a Custom deployment blade that pulls a small wrapper ARM template directly from this repository.

Click the Deploy CLMS button first. The blade prompts for resource group, region, VM name, admin username and admin password. The deployment takes about 8 to 10 minutes; the VM then needs a further 15 minutes to finish initialising before the UI is reachable.

Once CLMS is up, click the Deploy vPB button. Same blade pattern, different image. Use the same resource group as CLMS so both VMs end up in one tidy place that you can delete cleanly later.

Open the CLMS UI, create a project, copy the project key, and run `quickstart.sh` from a terminal (or Cloud Shell) to deploy sensors. The Portal does not yet have a click-through for sensors because they touch every tagged VM in the subscription; that is intentionally an explicit step.

Marketplace listing names

In the Portal search, CLMS appears as 'CloudLens Manager (Preview)' and vPB appears as 'CloudLens Virtual Packet Broker'. The Deploy buttons on the site point at the correct offer IDs automatically, so you do not need to find them by hand.

No screenshots, by design

Azure Portal UI shifts every few months. Rather than ship stale screenshots, this guide describes the click path in words. The actual buttons and field names match the Portal at June 2026.

7. Verification Checklist

Walk this list after every deploy. If any item fails, jump to section 8 for the matching troubleshooting entry.

- ☐ CLMS UI reachable on `https://<clms-public-ip>/`. The page loads with the Keysight CloudLens login screen, not a connection error.
- ☐ Logged in to CLMS with the default credentials (admin / Cl0udLens@dm!n). Default password reset prompt appears; set a strong password before going further.
- ☐ Project created in CLMS under Settings > Projects, and the API key has been copied into a safe place.
- ☐ vPB SSH reachable on port 22 of the vPB management public IP after 10 to 15 minutes. Login as admin with the admin password set during deploy.
- ☐ vPB CLI prompts confirm 'CloudLens vPB Console' on first SSH; reach config mode with 'enable' then 'configure'.
- ☐ Sensors visible in CLMS under Sensors page within 8 minutes of running `quickstart.sh`. Each sensor shows its hostname and Connected status.
- ☐ Test traffic from a sensed VM appears at the destination tool or probe (CLMS Investigate page or your downstream collector).

Printable

This page is intentionally a single checklist so it can be printed and ticked off during a customer handover or installation walkthrough.

8. Troubleshooting Reference

Organised by failure mode. The full reference (every error message we have seen in the field, plus the fix) lives at docs/TROUBLESHOOTING.md in the repository.

Marketplace terms

Symptom	Likely cause	Fix
MarketplacePurchaseEligibilityFailed during deploy	Marketplace terms not accepted for the publisher / offer	az vm image terms accept --publisher keysight-technologies-cloudlens --offer <offer> --plan <plan>
Deploy succeeds but VM never starts	Plan name typo (case-sensitive)	Use exact plan ids: clms-6-13-0_76 and cloudlens-virtual-packet-broker-3-15-0_1

Quota

Symptom	Likely cause	Fix
QuotaExceeded on DSv5 family	Subscription default quota is too low	Request quota increase in the Portal: Subscriptions > Usage + quotas; or pick a region with headroom
Quota probe warns but deploy continues	Quota is tight but enough for one VM	Safe to ignore; the script does not block on quota warnings

Network

Symptom	Likely cause	Fix
CLMS UI not reachable after 20 minutes	NSG blocks port 443 from your client IP	az network nsg rule create to allow your IP, or use Bastion
vPB SSH times out	NSG blocks port 22 from your client IP	Add an NSG rule for your client IP on port 22
Sensor cannot reach CLMS	Egress blocked from VM subnet to CLMS public IP on 443	Open egress on the workload subnet; or peer the VNets

Timing

Symptom	Likely cause	Fix
CLMS UI returns 502 / 503 just after deploy	CLMS is still initialising (services start in sequence)	Wait the full 15 minutes; the script's poll loop handles this

vPB SSH connection refused immediately after deploy	vPB OS is still booting and applying first-run config	Wait 10 minutes; if still refused, restart the VM from the Portal
---	---	---

9. Cleanup / Decommission

Two ways to undo a stack deploy. Pick the one that matches how you deployed it.

Terraform deploy

If you used Path B (Terraform), use terraform destroy. This removes exactly what Terraform created and leaves anything else in the resource group untouched.

```
cd deploy/terraform/stack
terraform destroy
```

Bash or Portal deploy

If you used Path A or Path C, the simplest cleanup is to delete the whole resource group. This is irreversible; double-check the RG name before pressing enter.

```
az group delete --name cloudlens-rg --yes --no-wait
```

The --no-wait flag returns immediately. The actual delete happens in the background and finishes in 1 to 3 minutes.

Sensors on workload VMs

Sensors are not destroyed by terraform destroy or az group delete, because they live on the customer's workload VMs (which are usually in a different resource group). To remove them:

- Linux: docker stop cloudlens-agent && docker rm cloudlens-agent
- Windows: Stop-Service CloudLensAgent; sc.exe delete CloudLensAgent
- Or: untag the VMs (remove cloudlens=yes) and rerun the playbook with --tags remove

Marketplace billing

Deleting the resource group also stops the Keysight Marketplace subscription billing for those VMs at the next hourly tick. Compute billing stops immediately. Allow up to 24 hours for the Marketplace billing line to disappear from your invoice.

10. Appendix: File Paths and Quick Links

Repository file paths

File	Purpose
deploy/deploy-stack.sh	Path A entry point (Bash one-liner)
deploy/terraform/stack/	Path B entry point (Terraform module)
deploy/terraform/stack/terraform.tfvars.example	Path B sample input
deploy/clms-marketplace.json	Wrapper ARM template for CLMS (used by Paths A and C)
deploy/vpb-marketplace.json	Wrapper ARM template for vPB (used by Paths A and C)
docs/CloudLens_Stack_Deployment_Runbook.pdf	This document (PDF)
docs/CloudLens_Stack_Deployment_Runbook.docx	This document (Word)
quickstart.sh	Sensor deployment chain (called by deploy-stack.sh)

Command reference

Action	Command
Run full stack (recommended)	<code>curl -sSL https://raw.githubusercontent.com/Keysight-Tech/cloudlens-ansible-azure/main/deploy/deploy-stack.sh bash</code>
Run full stack (dry-run)	<code>bash deploy/deploy-stack.sh --dry-run</code>
Run CLMS only (skip vPB)	<code>bash deploy/deploy-stack.sh --no-vpb</code>
Terraform apply	<code>cd deploy/terraform/stack && terraform apply</code>
Terraform destroy	<code>cd deploy/terraform/stack && terraform destroy</code>
Delete everything (RG)	<code>az group delete -n cloudlens-rg --yes --no-wait</code>

Useful links

Resource	Location
GitHub repository	https://github.com/Keysight-Tech/cloudlens-ansible-azure
Landing page	https://keysight-tech.github.io/cloudlens-ansible-azure/
Sensor runbook (companion document)	<code>docs/CloudLens_Ansible_Azure_Customer_Runbook.pdf</code>
Deployment guide (technical)	<code>docs/DEPLOYMENT_GUIDE.md</code>
Troubleshooting reference	<code>docs/TROUBLESHOOTING.md</code>
Issues / support	https://github.com/Keysight-Tech/cloudlens-ansible-azure

	azure/issues
Related: vPB + Azure GWLB	https://github.com/Keysight-Tech/cloudlens-vpb-azure-gwlb

Support email subject (for the Keysight account team): 'CloudLens Stack: <customer name>, <brief issue>'.